Discontinued
Last time buy: August 31, 2012

Panasonic
ideas for life

**Input/output relays for interface between the CPU and the load**

# I/O-RELAYS



**RoHS compliant**

## FEATURES

1. **Input and output modules for interfacing between CPU and external input devices or loads**
   - **IAC: AC input (yellow)**
   - **OAC: AC output (black)**
   - **IDC: DC input (white)**
   - **ODC: DC output (red)**
2. **Excellent transient noise immunity**
3. **Breakdown voltage: 4,000 V between input and output**
4. **Zero-cross switching for reducing EMI**
5. **LED operation indication types available**

## TYPICAL APPLICATIONS

1. **Optimum for minicomputer and microcomputer use, and for feed back control process control systems.**
2. **Robot, NC machine, Automatic Assembling machine.**
3. **Programmable controllers controls for motors, solenoids and solenoid valves.**

## ORDERING INFORMATION

□ □ – □ □ – □

Input or Output
I: Input module
O: Output module

AC or DC
AC: AC input or AC output
DC: DC input or DC output

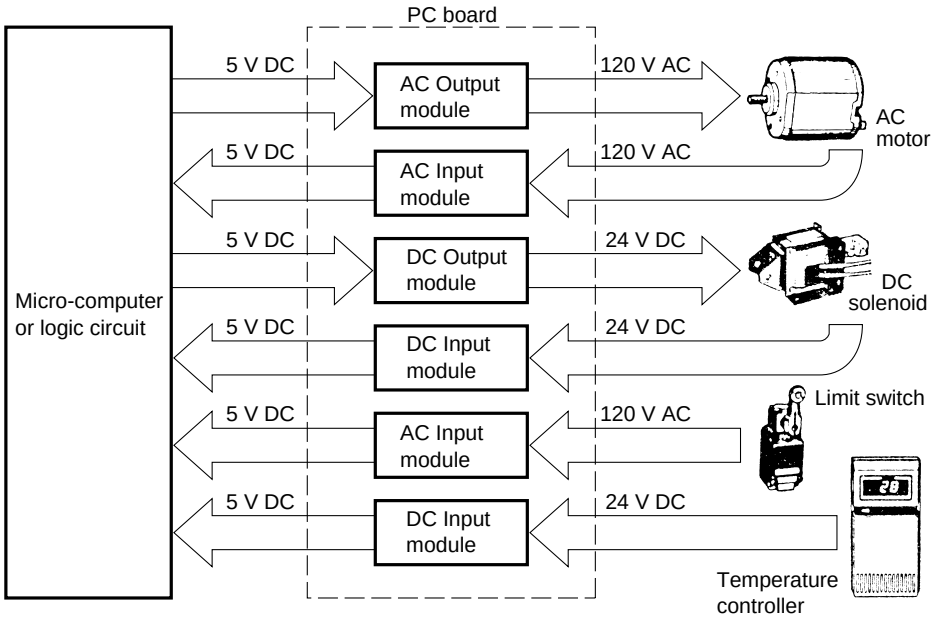
Logic supply or control voltage
3/15: 3 to 15 V DC
4/15: 4 to 15 V DC
10/32: 10 to 32 V DC

Input or Output Voltage
Input
Nil: 80 to 140 V AC
3 to 32 C DC
A: 160 to 280 V AC
Output
Nil: 75 to 125 V AC
3 to 60 V DC
A: 75 to 250 V AC
10 to 200 V DC

Operation indication
Nil: No operation indication
L: Operation indication

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# EXAMPLE OF I/O SYSTEM CONSTRUCTION




Recently, the microcomputer shown in the left has come into use, with a rapid increase in the use of feedback control for process controllers. As an input/output interface relay, it represents a fast responding interface with long life, noiseless, PC board mounting facility, making circuit design and manufacture simple, and providing amplification of maintenance with space saving. Our company's I/O relays have been grouped for use with either AC or DC devices as AC output type, AC input type, DC output type, and DC input. In addition, the I/O use relay is modularized identically, responding to the various load types for PC board mounting. Because combinations can be freely used, circuit design has been greatly facilitated. Accordingly, the above diagram of interface PC board mounting is a practical construction which can be used.

# TYPES

### 1. Input modules

| Type | Logic supply voltage | Input voltage | Part No. | |
|---|---|---|---|---|
| | | | No LED | With LED |
| AC | 4 to 15 V DC | 80 to 140 V AC | IAC4/15 | IAC4/15-L |
| | | 160 to 280 V AC | IAC4/15-A | IAC4/15-AL |
| | 10 to 32 V DC | 80 to 140 V AC | IAC10/32 | IAC10/32-L |
| | | 160 to 280 V AC | IAC 10/32-A | IAC10/32-AL |
| DC | 4 to 15 V DC | 3 to 32 V DC | IDC4/15 | IDC4/15-L |
| | 10 to 32 V DC | 3 to 32 V DC | IDC10/32 | IDC10/32-L |

Standard packing: Carton: 20 pcs.; Case: 200 pcs.
Note: Please refer to the Standards Chart for products that have been certified.

### 2. Output modules

| Type | Logic supply voltage | Output voltage | Part No. | |
|---|---|---|---|---|
| | | | No LED | With LED |
| AC | 3 to 15 V DC or 4 to 15 V DC | 75 to 125 V AC | OAC3/15 | OAC4/15-L |
| | | 75 to 250 V AC | OAC3/15-A | OAC4/15-AL |
| | 10 to 32 V DC | 75 to 125 V AC | OAC10/32 | OAC10/32-L |
| | | 75 to 250 V AC | OAC10/32-A | OAC10/32-AL |
| DC | 3 to 15 V DC or 4 to 15 V DC | 3 to 60 V DC | ODC3/15 | ODC4/15-L |
| | | 10 to 200 V DC | ODC3/15-A | ODC4/15-AL |
| | 10 to 32 V DC | 3 to 60 V DC | ODC10/32 | ODC10/32-L |
| | | 10 to 200 V DC | ODC10/32-A | ODC10/32-AL |

Standard packing: Carton: 20 pcs.; Case: 200 pcs.
Note: Please refer to the Standards Chart for products that have been certified.

# RATING

**1. Rating** (Ambient temperature: 20°C 68°F, Voltage ripple of logic side: 1% or less)

1) Input module

| Item | Part No. | IAC 4/15 | IAC 4/15-L | IAC 10/32 | IAC 10/32-L | IAC 4/15-A | IAC 4/15-AL | IAC 10/32-A | IAC 10/32-AL | IDC 4/15 | IDC 4/15-L | IDC 10/32 | IDC 10/32-L | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Input side | Input voltage | 80 to 140 V AC | | | | 160 to 280 V AC | | | | 3 to 32 V DC | | | | |
| | Input current, Max. | 5 mA (100 V) | | | | 5 mA (200 V) | | | | 20 mA (32 V) | | | | |
| | Operate voltage, Max. | 80 V AC | | | | 160 V AC | | | | 3 V | | | | |
| | Release voltage Min. | 10 V AC | | | | 20 V AC | | | | 0.8 V | | | | |
| Logic side | Logic supply voltage | 4 to 15 V | | 10 to 32 V | | 4 to 15 V | | 10 to 32 V | | 4 to 15 V | | 10 to 32 V | | |
| | Logic supply current, Max. | 15 mA | | | | | | | | 15 mA | | | | at max. logic supply voltage |
| | Breakdown voltage | 30 V | | | | | | | | 30 V | | | | |
| | Output current, Max. | 25 mA | | | | | | | | 25 mA | | | | |
| | Max. "OFF-side" leakage current | 100 μA | | | | | | | | 100 μA | | | | |
| | Max. "ON-side" voltage drop | 0.4 V | | | | | | | | 0.4 V | | | | at max. carrying current |

2) Output module

| Item | Part No. | OAC 3/15 | OAC 4/15-L | OAC 10/32 | OAC 10/32-L | OAC 3/15-A | OAC 4/15-AL | OAC 10/32-A | OAC 10/32-AL | ODC 3/15 | ODC 4/15-L | ODC 10/32 | ODC 10/32-L | ODC 3/15-A | ODC 4/15-AL | ODC 10/32-A | ODC 10/32-AL | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Logic side | Logic voltage range, DC | 3 to 15 V | 4 to 15 V | 10 to 32 V | | 3 to 15 V | 4 to 15 V | 10 to 32 V | | 3 to 15 V | 4 to 15 V | 10 to 32 V | | 3 to 15 V | 4 to 15 V | 10 to 32 V | | *1 |
| | Input Impedance (Approx.) | 1.6 kΩ | 1.7 kΩ | 6 kΩ | 5 kΩ | 1.6 kΩ | 1.7 kΩ | 6 kΩ | 5 kΩ | 1.6 kΩ | 1.7 kΩ | 6 kΩ | 5 kΩ | 1.6 kΩ | 1.7 kΩ | 6 kΩ | 5 kΩ | |
| | Drop-out voltage, Min. | 0.8V | | | | | | | | 0.8V | | | | | | | | |
| Output side | Max. load current | 2 A (Ambient temperature: Max. 30°C)*2 | | | | | | | | 2 A (Ambient temperature: Max. 40°C)*2 | | | | 1 A (Ambient temperature: Max. 40°C)*2 | | | | |
| | Output voltage | 75 to 125 V AC | | | | 75 to 250 V AC | | | | 3 to 60 V DC | | | | 10 to 200 V DC | | | | |
| | Non-repetitive surge current | 30 A (in one cycle at 60 Hz)*3 | | | | | | | | 5 A (for 1 s)*3 | | | | | | | | |
| | Max. "OFF-state" leakage current | 5 mA (100 V) | | | | 5 mA (200 V) | | | | 1 mA (60 V) | | | | 1 mA (200 V) | | | | at 60 Hz |
| | Max. "ON-state" voltage current | 1.6 V | | | | | | | | 1.6 V | | | | 2.3 V | | | | at max. carrying current |
| | Min. load current | 50 mA*4 | | | | | | | | 5 mA*4 | | | | | | | | |

Notes: *1. Please refer to REFERENCE DATA, "3. Input current vs. input voltage".
*2. Please refer to REFERENCE DATA, "1. Load current vs. ambient temperature".
*3. Please refer to REFERENCE DATA, "2. Surge current vs. time".
*4. When the load current is less than the rated minimum load current, please refer to "Cautions for Use of SSR".

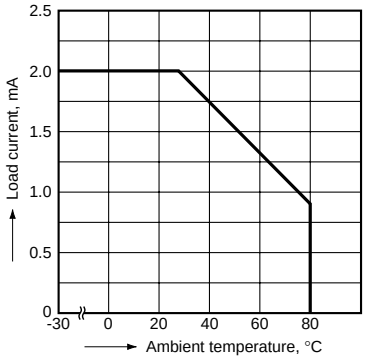
**2. Characteristics** (Ambient temperature: 20°C 68°F, Voltage ripple of logic side: 1% or less)

| | | Input modules | | Output modules | |
|---|---|---|---|---|---|
| | | IAC | IDC | OAC | ODC |
| Operate time, max. | | 20 ms | 5 ms | 1/2 cycle of voltage sine wave + 1 ms | 0.5 ms |
| Release time, max. | | 20 ms | 5 ms | 1/2 cycle of voltage sine wave + 1 ms | 2 ms |
| Insulation resistance | | $10^9$ Ω between input and output (at 500 V DC) | | | |
| Breakdown voltage | | 4,000 V AC between input and output (for 1 min.) | | | |
| Vibration resistance | Destructive | 10 to 55 Hz at double amplitude of 2 mm (1hour for X, Y, Z axis) | | | |
| | Functional | 10 to 55 Hz at double amplitude of 2 mm (10 minutes for X, Y, Z axis) | | | |
| Shock resistance | Destructive | Min. 980 m/s² {100G} (5 times each for X, Y, Z axis) | | | |
| | Functional | Min. 980 m/s² {100G} (4 times each for X, Y, Z axis) | | | |
| Ambient temperature | | –30°C to +80°C –22°F to +176°F | | | |
| Storage temperature | | –30°C to +100°C –22°F to +212°F | | | |
| Operational method | | – | | Zero-cross (Turn-ON and Turn-OFF) | – |

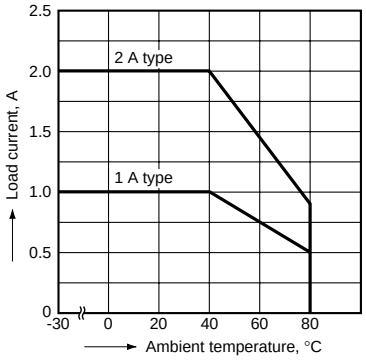
**Discontinued**
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# REFERENCE DATA

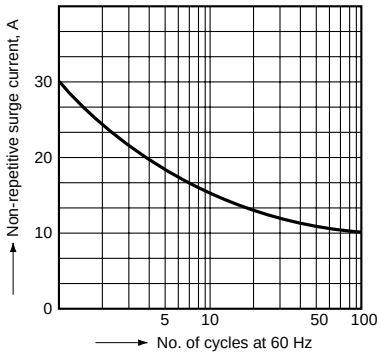
1-1. Load current vs. ambient temperature characteristics (OAC)



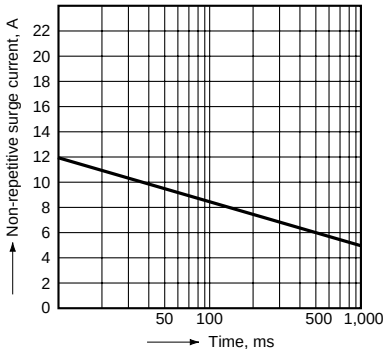

1-2. Load current vs. ambient temperature characteristics (ODC)



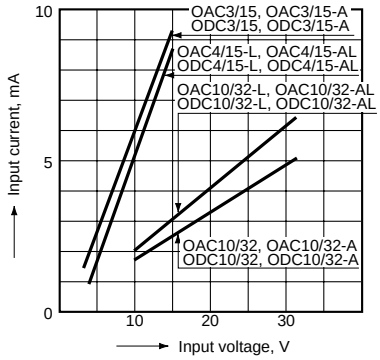

2-1. Non-repetitive surge current vs. time characteristics (OAC)




2-2. Non-repetitive surge current vs. time characteristics (ODC)




3. Input current vs. input voltage characteristics (OAC, ODC)




# DIMENSIONS (mm inch)

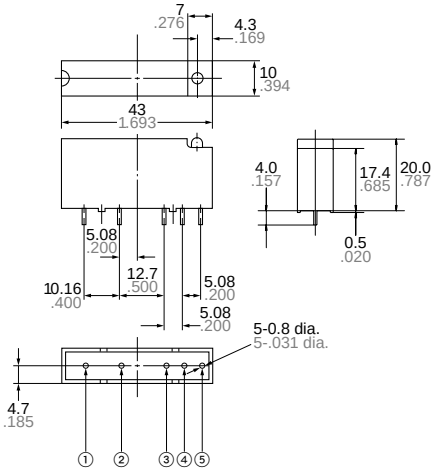
The CAD data of the products with a **CAD Data** mark can be downloaded from: http://industrial.panasonic.com/ac/e

## 1. Input module (IAC, IDC)

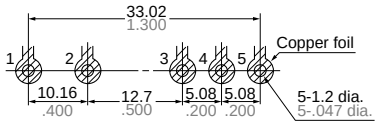
**CAD Data**



External dimensions




PC board pattern (Bottom view)



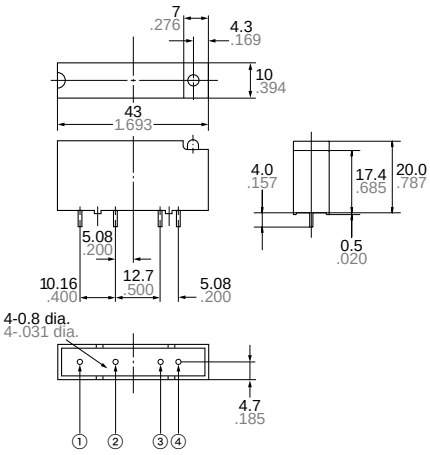

| | IAC | IDC |
|---|---|---|
| 1 | Input: AC | Input: DC + |
| 2 | Input: AC | Input: DC – |
| 3 | Vcc | Vcc |
| 4 | Logic output | Logic output |
| 5 | Grounding | Grounding |
| Body color | Yellow | White |

## 2. Output modules (OAC, ODC)

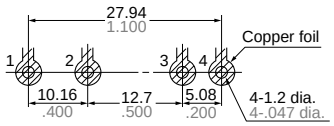
**CAD Data**



External dimensions




PC board pattern (Bottom view)

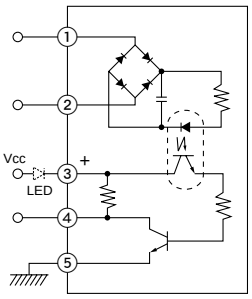



| | OAC | ODC |
|---|---|---|
| 1 | Output: AC | Output: DC + |
| 2 | Output: AC | Output: DC – |
| 3 | Input: DC+ | Input: DC+ |
| 4 | Input: DC- | Input: DC- |
| Body color | Black | Red |

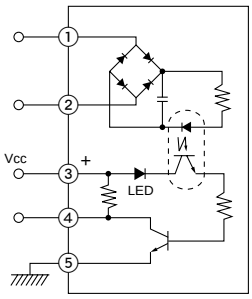
# BLOCK DIAGRAM

### 1) AC input module

Without LED

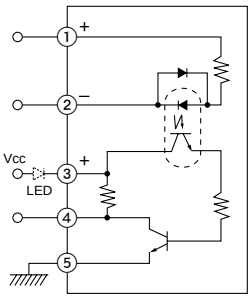



With LED

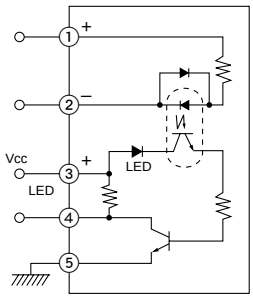



### 2) DC input module

Without LED

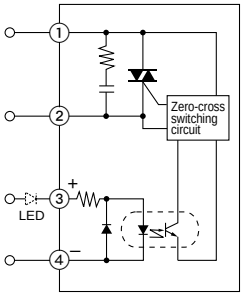



With LED

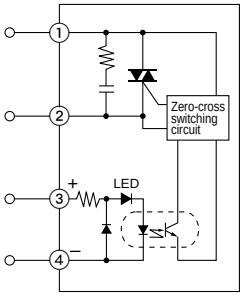



### 3) AC output module

Without LED

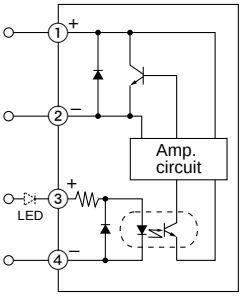



With LED

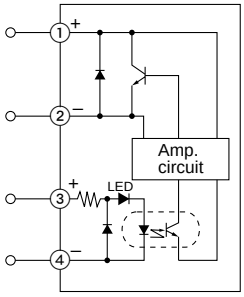



### 4) DC output module

Without LED




With LED



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# ACCESSORIES (mm inch)

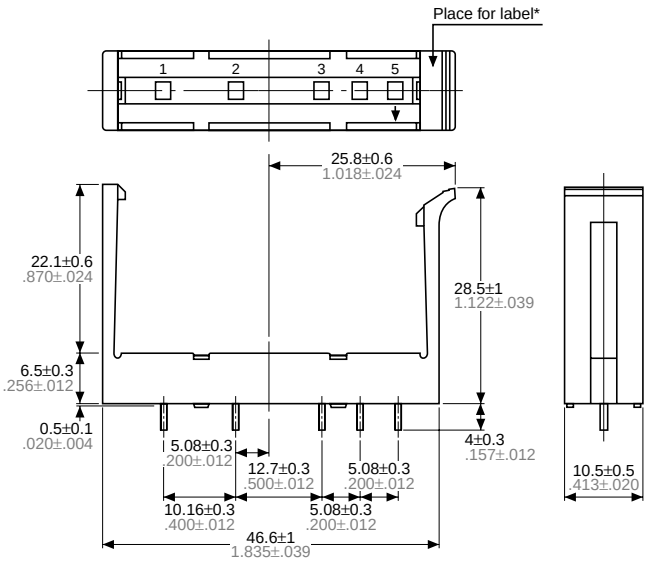
## I/O Socket

CAD Data

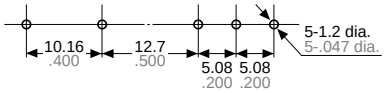


I/O-PS

External dimensions




PC board pattern (Bottom view)




* Use the included label. Affixing it here on the socket will facilitate viewing of the circuit and other tasks.

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Panasonic:

IAC4/15 IAC4/15-AL IAC4/15-L IDC4/15-L OAC10/32-AL OAC3/15 OAC3/15-A OAC4/15-AL OAC4/15-L ODC10/32-A ODC10/32-L ODC3/15 ODC3/15-A ODC4/15-L